

Complete Thesis Structure

Title of the Thesis

[Insert full thesis title]

A Thesis Submitted to

[Department / Institute / University Name]

In Partial Fulfillment of the Requirements for the Degree of

[Degree Name]

Submitted by

[Student Name]

[Student ID / Registration No.]

Supervised by

[Supervisor Name, Designation, Affiliation]

[Co-supervisor / Joint Supervisor, if applicable]

[Department Name]

[University Name]

[City, Country]

[Month, Year]

Approval Page

This thesis entitled "[Insert Thesis Title]" submitted by [Student Name] has been accepted as satisfactory in partial fulfillment of the requirements for the degree of [Degree Name].

Board of Examiners / Thesis Committee

Chairman: [Name, Designation]

Supervisor: [Name, Designation]

Member: [Name, Designation]

External Member: [Name, Designation]

Declaration

I hereby declare that this thesis is my own original work and has not been submitted elsewhere for any degree or diploma, except where due acknowledgment has been made in the text.

Signature

[Student Name]

[Date]

Certificate / Supervisor's Certification

This is to certify that the thesis entitled "[Insert Thesis Title]" has been carried out by [Student Name] under my/our supervision and is a record of original research work.

Supervisor Signature

[Name]

[Designation]

[Date]

Acknowledgments

[Thank supervisor, department, funding agency, lab members, collaborators, friends, and family.]

Abstract

[One-page summary of the problem, background, objectives, methods, main results, and conclusion.]

Keywords: [keyword 1], [keyword 2], [keyword 3], [keyword 4], [keyword 5]

Table of Contents

[Auto-generated]

List of Figures

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List of Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

DL – Deep Learning

CNN – Convolutional Neural Network

CI – Computational Intelligence

[Add as needed]

List of Symbols (if applicable)

(x) = [meaning]

(y) = [meaning]

(α) = [meaning]

Chapter 1: Introduction

[Chapter Abstract; Points Discussed in the Chapter]

1.1 Background of the Study

[Introduce the broad research area.]

[Explain why the field is important.]

[Narrow toward the specific topic.]

1.2 Problem Statement

[Clearly state the research problem.]

[Explain why it is important and what consequences it has.]

1.3 Research Gap

[Discuss what previous studies have done.]

[Identify what remains unresolved, weak, or missing.]

1.4 Research Aim

[State the overall aim of the thesis.]

1.5 Research Objectives

- Objective 1: [Insert objective]
- Objective 2: [Insert objective]
- Objective 3: [Insert objective]

1.6 Research Questions / Hypotheses

Research Questions

- RQ1: [Insert question]
- RQ2: [Insert question]
- RQ3: [Insert question]

or

- Hypotheses
- H1: [Insert hypothesis]
- H2: [Insert hypothesis]

1.7 Significance of the Study

[Explain theoretical, methodological, practical, industrial, social, or policy significance.]

1.8 Scope of the Study

[Define what is included and what is excluded.]

1.9 Limitations of the Study

[State expected limitations such as dataset size, equipment constraints, time, or generalizability.]

1.10 Thesis Organization

This thesis is organized as follows:

- Chapter 1 presents the introduction.
- Chapter 2 discusses the problem domain, identifies ML/CI prospects, and reviews the relevant literature.
- Chapter 3 describes the methodology.
- Chapter 4 presents the experimental studies and analysis.
- Chapter 5 discusses engineering, ethical, and professional considerations.
- Chapter 6 explains the complex engineering problems and activities associated with the project.
- Chapter 7 concludes the thesis and suggests future work.

Chapter 2: Literature Review

[Chapter Abstract; Points Discussed in the Chapter]

2.1 Background of the Problem

[2–3 pages with related figures, if necessary.]

[Provide basic knowledge of the problem in detail.]

[Define the important theories, concepts, and technical foundations.]

Example prompts:

- Explain Sundarban, local birds, and the importance of avian biodiversity.
- Explain cancer, its effects on human life, and the importance of early detection.

2.2 Significance of ML or CI in [Problem Domain]

[Explain the importance of ML or CI in addressing the selected problem.]

2.2.1 ML/DL Basics

[Introduce the basic concepts of ML and DL.]

2.2.2 ML/DL for Automatic Detection

[Explain ML/DL-based automatic detection with a suitable figure.]

2.2.3 Benefits of ML-Based Detection

[Discuss the benefits of ML-based detection.]

2.2.4 Data Requirements for Training ML Models

[Explain the amount, type, and quality of data needed to train a reliable model.]

2.3 Benchmark Dataset or Procedure of Data Collection

[Describe the benchmark dataset, or explain the procedure of data collection.]

2.4 Review of Previous Studies

[Discuss related studies in thematic or chronological order.]

2.4.1 Studies Based on Conventional ML Methods

[Review prior work.]

Example: Solutions with conventional ML-based methods.

2.4.2 Studies Based on DL Methods

[Review prior work.]

Example: Solutions with DL-based methods.

2.4.3 Studies Based on Advanced / Hybrid Methods

[Review prior work.]

Example: Solutions with advanced or hybrid methods.

2.5 Summary of Existing Methods / Models / Approaches

[Compare methodologies, datasets, algorithms, systems, and analytical approaches.]

2.6 Critical Analysis of Previous Work

[Point out weaknesses, inconsistencies, gaps, limited datasets, lack of validation, poor generalization, or methodological limitations.]

2.7 Identified Research Gap

[Summarize the exact gap your thesis addresses.]

2.8 Conceptual Framework / Proposed Research Direction

[Describe the conceptual model, framework, or direction that motivates your study.]

2.9 Chapter Summary

[Briefly summarize the chapter and link to the methodology chapter.]

Chapter 3: Research Methodology

[Chapter Abstract; Points Discussed in the Chapter]

3.1 Research Framework

[Explain the overall method framework with a clear figure.]

[The following subsections will describe the individual parts of the framework.]

3.2 Data Collection / Sample Preparation

[Explain how data or samples were collected or prepared.]

3.3 Data Preprocessing and Feature Generation

[Describe data preprocessing, cleaning, normalization, augmentation, segmentation, or feature generation.]

3.4 Algorithm / Model Development

[Describe the algorithm, model, or processing steps.]

3.5 Experimental Design / Validation Strategy

[Explain train-test split, cross-validation, benchmarking, simulation settings, or validation procedure.]

3.6 Evaluation Metrics / Performance Measuring Techniques

[Explain how the model's proficiency will be evaluated.]

[Describe statistical tests, significance level, confidence intervals, model evaluation metrics, error measures, or performance indicators.]

3.7 Novelty and Significance of the Proposed Method

[Identify the novelty and significance of the proposed method compared with the state of the art.]

Chapter 4: Experimental Studies

[Chapter Abstract; Points Discussed in the Chapter]

4.1 Experimental Environment

[Describe hardware, software versions, operating system, platforms, instruments, lab conditions, or dataset partitions.]

4.2 Parameter Settings

[Describe hyperparameters, thresholds, simulation parameters, and configuration settings.]

4.3 Baseline / Comparative Methods

[List the methods or systems used for comparison.]

4.4 Procedure of Experimentation

[Provide a step-by-step explanation of the experiment execution.]

4.5 Experimental Results

[Provide findings with tables, graphs, and figures.]

4.6 Experimental Analysis / Ablation Study

[Analyze the contribution of individual components or settings.]

4.7 Comparative Analysis

[Compare your method/results with baselines, previous studies, or alternative settings.]

4.8 Discussion of Findings

[Interpret the results and explain their meaning.]

4.9 Chapter Summary

[Summarize the key findings.]

Chapter 5: Engineering, Ethical, and Professional Considerations

[Chapter Abstract; Points Discussed in the Chapter]

5.1 Introduction

[Introduce the chapter and explain why these considerations are important in the context of the proposed work.]

5.2 Intellectual Property Considerations

[Discuss ownership, authorship, copyright, patent issues, software licensing, dataset usage rights, and citation responsibilities.]

5.3 Ethical Considerations

[Discuss fairness, bias, responsible research conduct, informed consent, transparency, explainability, and possible misuse.]

5.4 Data Privacy and Security Considerations

[Explain privacy, confidentiality, anonymization, secure storage, secure transmission, access control, and protection of sensitive data.]

5.5 Safety Considerations

[Discuss safety issues related to deployment, system misuse, inaccurate prediction, false positives, false negatives, or operational risks.]

5.6 Legal Considerations

[Explain legal and regulatory compliance issues, including data regulations, institutional requirements, licensing, and domain-specific legal concerns.]

5.7 Societal, Health, and Cultural Impact

[Discuss the positive and negative impacts of the work on society, health, communities, and cultural context.]

5.8 Environmental and Sustainability Considerations

[Discuss energy consumption, computational cost, sustainability, efficient resource use, and environmental relevance.]

5.9 Summary

[Summarize the key engineering, ethical, legal, and professional considerations.]

Chapter 6: Complex Engineering Problems and Activities

[Chapter Abstract; Points Discussed in the Chapter]

6.1 Introduction

[Introduce the concept of complex engineering problems and activities in relation to the project.]

6.2 Complex Engineering Problems Associated with the Project

[Discuss the complex engineering problems addressed by the project.]

Examples: high-dimensional data, uncertainty, noisy data, limited samples, domain constraints, conflicting requirements, interpretability, and optimization difficulty.

6.3 Complex Engineering Activities Associated with the Project

[Discuss the engineering activities performed.]

Examples: system design, model development, feature engineering, experimentation, validation, optimization, comparative analysis, and problem-solving under constraints.

6.4 Mapping of the Proposed Work with Complex Engineering Attributes

[Map the thesis work with recognized complex engineering attributes.]

Possible mapping items include:

- Requirement of deep engineering knowledge
- Conflicting technical and practical constraints
- Use of abstraction and modeling
- Presence of uncertainty and incomplete information
- Need for investigation and experimentation
- Consideration of ethical, legal, social, health, safety, and environmental issues
- Use of modern engineering tools and interdisciplinary knowledge

[This section may be presented in a table.]

6.5 Summary

[Summarize how the proposed work addresses complex engineering problems and activities.]

Chapter 7: Conclusion and Future Work

[Chapter Abstract; Points Discussed in the Chapter]

7.1 Summary of the Study

[Restate the purpose, problem, and overall approach.]

7.2 Major Findings

[Summarize the major results.]

7.3 Limitations of the Study

[State the limitations of the study.]

7.4 Future Recommendation

[Provide recommendations for researchers, practitioners, policymakers, clinicians, engineers, or industry as appropriate.]

7.5 Closing Statement

[End with a strong final statement on the significance of the thesis.]

References

[Must be created Mendeley]

[Insert all cited works in the required style: APA, IEEE, Harvard, Vancouver, etc.]

Appendices

Appendix A: Questionnaire / Survey Instrument

[Insert questionnaire if applicable.]

Appendix B: Additional Tables

[Insert supplementary tables.]

Appendix C: Additional Figures

[Insert supplementary figures.]

Appendix D: Source Code / Pseudocode

[Insert code snippets or algorithmic details.]

Appendix E: Ethical Approval Letter

[Attach if applicable.]